MACHINE LEARING BASED IMAGE AND OBJECT RECOGNITION USING CV LENS

ABSTRACT

- This project is based on camera application.
- IMAGE module contains face emotions and image recognition.
- Face emotion detects a face and reveals one’s reactions such as happiness or sadness, etc.
- The process of OCR is to scan the photo for the text present in an image.
- This project is being developed using Android Studio with the Cognitive Services from Microsoft and Google API services.

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LITERATURE SURVEY

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FACE RECOGNITION USING EIGENFACES

- As eigenface is primarily a dimension reduction method.
- Its training process is completely automatic and easy to code.
- Raw data are directly used for learning and recognition without any processing.
- Eigenfaces recognition rate decreases for recognition under different pose & lighting.

LITERATURE SURVEY

HAAR-LIKE FEATURES FOR FAST FACE DETECTION

- Haar like features are extracted from gray level image**.**
- Haar + Adaboost schema used for fast face detection.
- Haar-like features can detect regions brighter or darker easily.
- Not so good for detecting profile faces.

LITERATURE SURVEY

FACIAL EXPRESSION RECOGNITION SYSTEM USING HOG

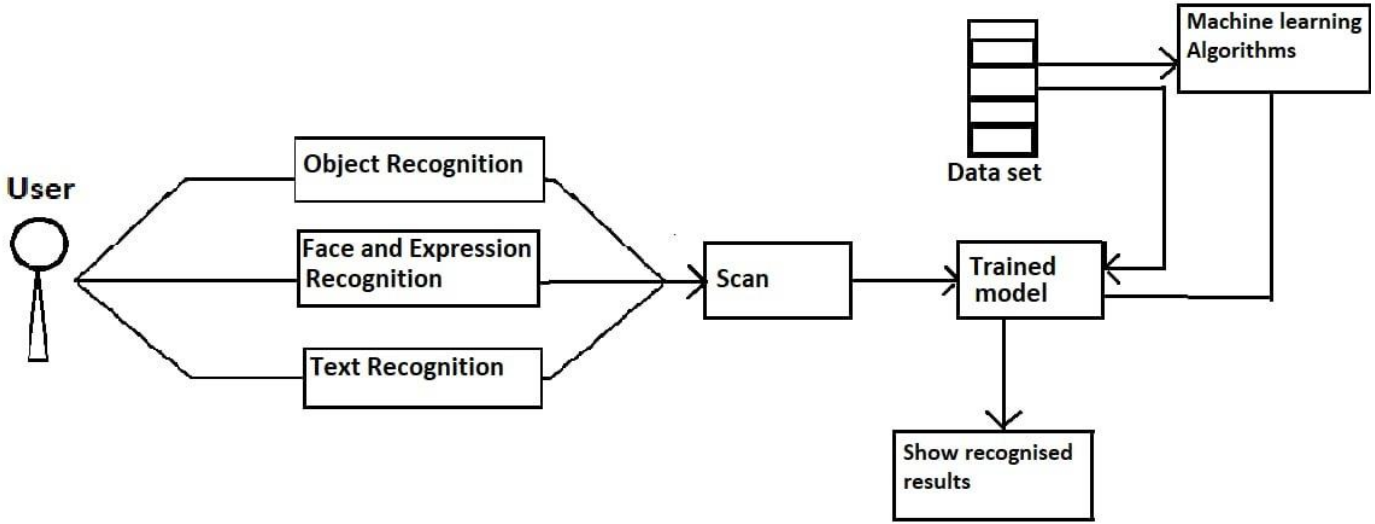
- Histogram of Oriented Gradients used to detect Faces and Objects in the Images.
- The FER process extracts HOG features from each facial region, and then, support vector machine is performed to classify the final facial expression.
- In smaller scales they provide more fine-grained detail.
- The final descriptor vector grows larger, thus taking more time to extract and to train using a given classifier.

LITERATURE SURVEY

CBP EMBEDDED WITH IMAGE EUCLIDEAN DISTANCE

- Improving the robustness to small perturbation (deformation) of expressional images.
- We introduce Image Euclidean Distance (IMED) and embed it in CBP.
- It decreases the white noise's influence on face images.
- Recognition accuracy is 90 percent on JAFFE database.

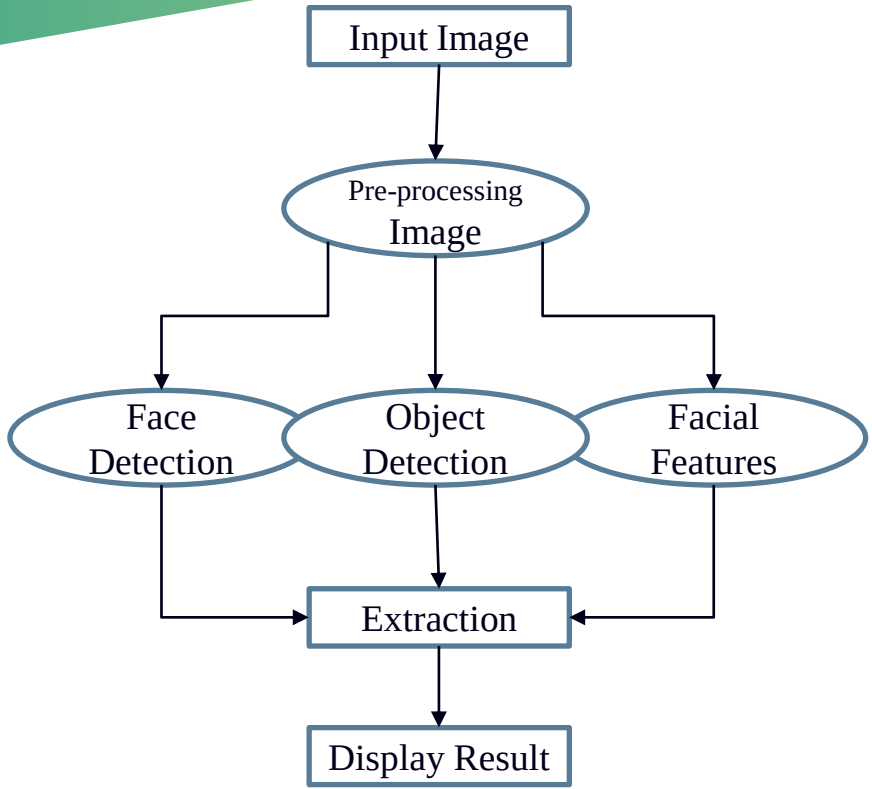
ARCHITECTURE



DATA FLOW DIAGRAM

- Data Flow Diagram shows how the user can access the system and how the result is obtained.
- It also shows the Image Processing and Image Extraction.

DATA FLOW DIAGRAM



MODULES IMPLEMENTATION

- **INPUT MODULE**: The input module is for the user to input the image or through camera lens to detect the objects.
- **USER:** User can access all the four modules.
  1. Object Detection
  2. Text Detection
  3. Face Recognition
- **PREDICTION SYSTEM**: It is the main processing system where the user's input is compared with the trained data and various algorithms predict the object or text based on the inputs given by the user.
- **OUTPUT MODULE**: This is the output which the user receives where it displays the predicted object name with rectangular boxes.

EXECUTION OUTPUT

- This is the output which the user receives where it displays the predicted object name with rectangular boxes.
- Text detection can be shown in words or single characters based on user choice.

DRAWBACKS

- System cannot differentiate between arts and real person.
- It cannot differentiate between real objects and virtual objects.
- If there is any camera error whole architecture will not work as basic input is taken from camera module.

CONCLUSION

- Today, one of the fields that uses facial recognition the most is security.
- Facial recognition is a very effective tool that can help law enforcers recognize criminals and software companies are leveraging the technology to help users access their technology.
- This can make other security measures such as passwords and keys obsolete.
- They are looking to leverage this technology to use faces as credit cards to pay for your transportation fee. Instead of having to go to a booth to buy a ticket for a fare, the face recognition would take your face, run it through a system, and charge the account that you've previously created